

SRRR FAILURE MEETING

Ritty/ Ferris



TOM 45 1

Pump Serial: ACS-1228

Model: 20-125-RHBC-16-5-4 BDV

Max Stroke: 167

Day Reported Down: 12/17/2025

Days Run: 292

Failure Cause: Pump Failure

Inspection Summary: Inspection revealed the plunger mic readings measured up to .007 and the barrel mic readings measured up to .006, indicating excessive wear. The balls and seats did not hold vacuum during testing, reflecting loss of sealing integrity and reduced pump efficiency.

WEAR (THOUSANDTHS INCH AT FEET FROM TOP)																																					
Feet	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
Prv BBL Out	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1																				
Barrel In	0	1	1	4	1	1	1	3	1	1	1	1	6	1	1	1	1																				
Barrel Out	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1										
Prv PLG Out	4	3	3	3	3	4																															
Plunger In	0	7	3	3	3	6																															
Plunger Out	0	5	5	5	5																																
Traveling Assembly		Partnumber							Description							Mtl Type	Part Condition		Qty	Replaced By																	
Coupling		S23N-OSI							2-1/2" with 3/4" pin 1/2" P.T. Stainless ,BUSHING, VALVE ROD							SS	GALLED		1	S27N-OSI - 2-1/2" for 7/8" Valve Rod - Stainless, Bushing, Collet Type																	
Valve Rod		114K-OSI							7/8" Valve Rod Stock, Per Foot							CS	WORN		25	114K-OSI																	
Top Plunger Adapter		M47D1-OSI							1-1/4" with 1/2" P.T. Monel,TPA							ML	GALLED		1	M47D1-OSI																	
Plunger		102DQ5-3-M-OSI							1-1/4" X 5', -.003 FIT, PLUNGER,M.P.							SM/M P	WORN		1	102DQ5-5-M-G-OSI- 1-1/4" X 5', -.005 FIT, PLUNGER,M.P. Grooved Body																	
Traveign Cage		M39D-OSI							1-1/4" Closed API PIN END MNL							ML	GALLED		1	M39D-OSI																	
Primary TV Ball		10DSN2-OSI							Ball, Silicon Nitrite - 11/16"							SN	WORN		1	10DSN2-OSI																	
Primary TV Seat		11DC-OSI							Seat, Tungsten Carbide - 1-1/4"							CAR	WORN		1	11DC-OSI																	
Seat Plug		S98D1-OSI							1-1/4" Recessed, Stainless, SEAT PLUG							SS	GALLED		1	S98D1-OSI																	
Standing Assembly		Partnumber							Description							Mtl Type	Part Condition		Qty	Replaced By																	
Valve Rod Guide		S70K1-V-OSI							2" for 7/8" Rod, Stainless - Vertical Discharge,GUIDE, ROD B							SS	WORN		1	S70K1-V-OSI																	
Barrel Bushing		S41K-M-OSI							Guide Mandrel, for Top Delrin Centralizer or Sand shield. 2"x15/16"							SS	WORN		1	S41K-M-OSI																	
Top Extension		B45D48-OSI							1-1/4" x 48" Brass,COUPLING, EXTENSION RH							BR																					
Barrel		B16D16N-OSI							16" Brass, Nickle Carbide,BARREL TH, 1-5/8" O.D. x 1-1/4" I.							BR/NC	WORN		1	B16D26N-OSI-26" Brass, Nickle Carbide,BARREL RH, 1-5/8" O.D. x 1-1/4" I.D.																	
Bottom Extension		B45D03-OSI							1-1/4" x 3", Brass,COUPLING, EXTENSION RH							BR	GALLED		1	B45D03-OSI																	
Primary SV		S37H-OSI							1-3/4" for 1-1/2" RW Tube, Stainless,Cage							SS	LIFE EXCEED		1	S39H1-OSI																	
Primary SV Ball		10HSN2-OSI							Ball, Silicon Nitride - 1"							SN	WORN		2	10HSN2-OSI																	
Primary SV Seat		11HC-OSI							Seat, Tungsten Carbide - 1-3/4"							CAR	WORN		2	11HC-OSI																	
Dbl SV Cage		S38H-OSI							1-3/4" Stainless,CAGE, CLOSED DOUBLE VALVE							SS	LIFE EXCEED		1	S38H-OSI																	
Hold Dow n Mandrel		S85K-OSI							2" API with 1.875 No Go, Stainless,MANDREL, API H.D., 3 CUP							SS	WORN		1	S85K-OSI																	
Spacer Rings		S110K-OSI							2" Type HR, Stainless, SPACER RING							SS	WORN		2	S110K-OSI																	
Seating cups		625J+30P-OSI							1-25/32" +30 #100, SEATING CUPS, TYPE "HR"							PLASTI	WORN		3	625J+30P-OSI																	
Lock Nut		S95K-OSI							2" Stainless LOCK NUT,CUP HOLD DOWN							SS	GALLED		1	S95K-OSI																	
Strainer Bushing		S50K-OSI							2" with 1" P.T. Stainless,COUPLING, HOLD DOWN							SS	GALLED		1	S50K-OSI																	
Collet Nut		S28N-OSI							2" & 2-1/2" Collet Nut- Stainless,Bushing							SS			1	S28N-OSI																	
Reason For Pulling: TUBING FAILURE																		Foreign Materials:																			
Pump Desc (After): 20-125-RHBC-26-5-0																		Accessories: CVS-DBL																			
Deliver To: Stock																		Max Stroke (After): 256																			
Plunger Fit (After): -5																		Valve Rod Size: 7/8"																			
Remark/Notes: CONVERTED PUMP TO SPECIFICATION FOR WELL BARREL WN PLUNGER WN PRIMARY TV BALL AND SEAT WN PRIMARY SV BALL AND SEAT WN PRIMARY SV CAGE LIFE																																					

TOM 45 1



1-1/4" X 5', -.005 FIT, PLUNGER,M.P Body, Worn.



Traveling cage, beat out

TOM 45 1



Slightly Tagged, Guide Rod, Vertical Discharge & Bushing.

TOM 45 1

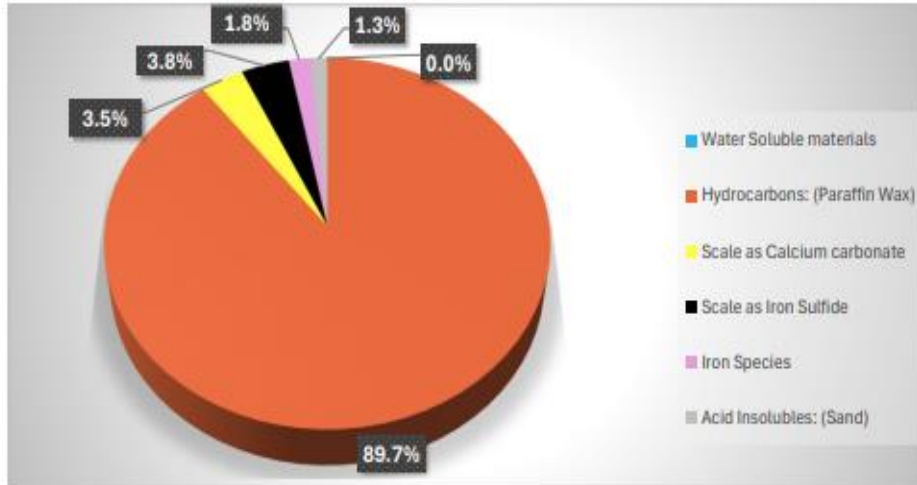


SOLID ANALYSIS TOM 45 1

SOLID COMPOSITION

Water Soluble materials	0.0%
Hydrocarbons: (Paraffin Wax)	89.7%
Scale as Calcium carbonate	3.5%
Scale as Iron Sulfide	3.8%
Iron Species	1.8%
Acid Insolubles: (Sand)	1.3%

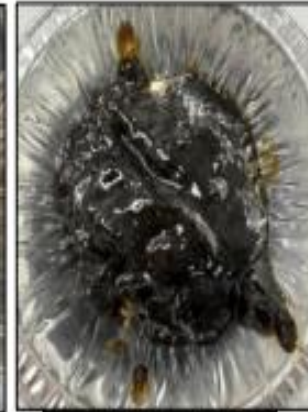
Solid Appearance: **Brown waxy solid**



SOLID APPEARANCE



Initial Sample



Isolated Paraffin Wax



Isolated Iron Species



After Cleaning (Microscopic View)

A high Paraffin wax content was detected and isolated, with a melting point of 60 – 63 °C. Small amounts of scale deposits composed of calcium carbonate and iron sulfide were also detected in the sample, with calcium carbonate confirmed by FTIR analysis.

Pump Serial: DBO-29 (CREATED NEW PUMP)
Model:20-125-RHBC-20-5
Day Reported Down: 12/17/2025
Days Run: 194
Failure Cause: Pump Failure

Inspection Summary: Inspection revealed the valve rod exhibited corrosion, and the valve rod bushing and collet nut were galled. The plunger was scored and worn, and the valve rod bushing showed tagging damage. The primary standing valve cage was galled to the secondary cage. The traveling valve ball and seat did not hold vacuum during testing, and the secondary standing valve ball and seat also failed vacuum testing.

WEAR (THOUSANDTHS INCH AT FEET FROM TOP)																																					
Feet	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
Prv BBL Out																																					
Barrel In	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2																
Barrel Out	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1																
Prv PLG Out																																					
Plunger In	6	6	5	5	7	7																															
Plunger Out	5	5	5	5	5	5																															
Traveling Assembly			Partnumber							Description											Mtl Type		Part Condition			Qty		Replaced By									
Valve Rod Bushing			S23N-OSI							2-1/2" with 3/4" pin 1/2" P.T. Stainless ,BUSHING, VALVE ROD											SS		TOOL MARKED/ GALLED			1											
Collet Nut																					SS		GALLED														
Valve Rod			114K-OSI							7/8" Valve Rod Stock, Per Foot											CS		CORROSION			25											
Top Plunger Adapter			M47D1-OSI							1-1/4" with 1/2" P.T. Monel,TPA											ML		GALLED			1											
Plunger			102DQ5-5-M-G-OSI							1-1/4" X 5', -.005 FIT, PLUNGER,M.P. Grooved Body											SM/MP		WORN / SAND CUT			1											
Primary Tv Cage			M39D-OSI							1-1/4" Closed API PIN END MNL											ML		LIFE EXCEED			1											
Primary TV Ball			10DSN2-OSI							Ball, Silicon Nitrite - 11/16"											SN		LEAKED			1											
Primary TV Seat			11DC-OSI							Seat, Tungsten Carbide - 1-1/4"											CAR		LEAKED			1											
Seat Plug			S98D1-OSI							1-1/4" Recessed, Stainless, SEAT PLUG											SS		REPLACED			1											
Standing Assembly			Partnumber							Description											Mtl Type		Part Condition			Qty		Replaced By									
Valve Rod Guide			S70K1-V-OSI							2" for 7/8" Rod, Stainless - Vertical Discharge,GUIDE, ROD B											SS		POUNDED			1											
Barrel Bushing			S41K-OSI							For 1-1/2" RW Barrel, Stainless ,CONNECTOR, UPPER BARREL											SS		GALLED			1											
Top Extension			B45D48-OSI							1-1/4" x 48" Brass,COUPLING, EXTENSION RH											BR		PART MISSING			1											
Barrel			B16D16N-OSI							16' Brass, Nickle Carbide,BARREL TH, 1-5/8" O.D. x 1-1/4" L											BR/NC		LIFE EXCEED			1											
Primary SV Cage			S39H1-OSI							1-3/4" Modified, 1-1/4" RH Barrel, Stainless,CAGE, CLOSED PI											SS		LIFE EXCEED / GALLED			1											
Primary SV Ball			10HSN2-OSI							Ball, Silicon Nitride - 1"											SN		STUCK IN GALLED CAGE			2											
Primary SV Seat			11HC-OSI							Seat, Tungsten Carbide - 1-3/4"											CAR		STUCK IN GALLED CAGE			2											
DbI SV Cage			S38H-OSI							1-3/4" Stainless,CAGE, CLOSED DOUBLE VALVE											SS		LIFE EXCEED			1											
Hold Down Mandrel			S85K-OSI							2" API with 1.875 No Go, Stainless,MANDREL, API H.D., 3 CUP											SS		GALLED			1											
Spacer Rings			S110K-OSI							2" Type HR, Stainless, SPACER RING											SS		STUCK ON HD MANDREL			2											
Seating cups			625J+30P-OSI							1-25/32" +30 #100, SEATING CUPS, TYPE "HR"											PLASTI		REPLACED			3											
Lock Nut			S95K-OSI							2" Stainless LOCK NUT,CUP HOLD DOWN											SS		GALLED			1											
Strainer Bushing			S50K-OSI							2" with 1" P.T. Stainless,COUPLING, HOLD DOWN											SS		TOOL MARKED			1											
Reason For Pulling: TUBING FAILURE																				Foreign Materials: Paraffin Sample taken																	
Pump Desc (After): 20-125-RHBC-16-5-4																				Accessories: CVS-DBL																	
Deliver To: Stock																				Max Stroke (After): 167																	
Plunger Fit (After):																				Valve Rod Size: 7/8"																	
Remark/Notes: Pump came in a 20' straight, replace w/ 4ft extension, v rod bushing collet nut were tool marked + galled, replaced w/single piece,vrod had corrosion, plunger was worn/ sand cut, tv cage was life exceeded, tv ball + seat leaked, vrod guide pounded, barrel bushing galled, top extension added, barrel was life exceeded, sv cages galled together, was able to remove only 1 set of balls + seats that leaked, sv cages were life exceeded, hold down mandrel + lock nut galled, strainer bushing was tool marked built new pump , new pump DBO-29																																					

TOM 4A 1



Plunger — Scored Surface.



Standing Valve — galled.



Exceeded recommended
Run Life cage.

TOM 4A 1



Lock Nut galled to the hold
down.



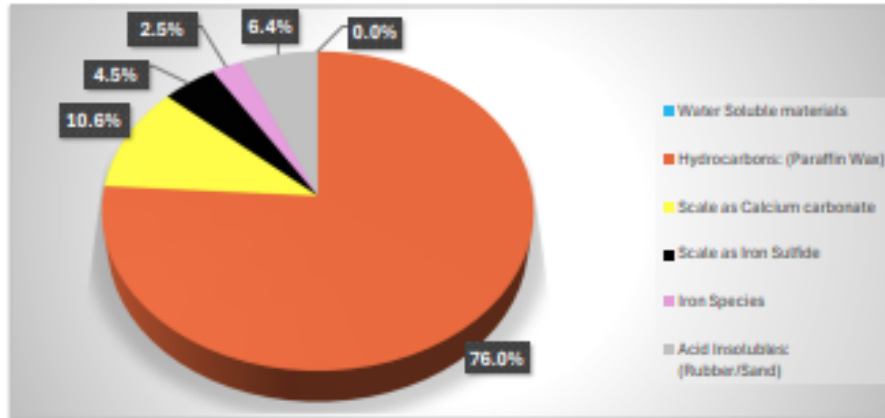
NO-GO – Worn

TOM 4A 1

SOLID COMPOSITION

Water Soluble materials	0.0%
Hydrocarbons: (Paraffin Wax)	76.0%
Scale as Calcium carbonate	10.6%
Scale as Iron Sulfide	4.5%
Iron Species	2.5%
Acid Insolubles: (Rubber/Sand)	6.4%

Solid Appearance: Brown waxy solid



SOLID APPEARANCE



Initial Sample



Isolated Paraffin Wax



Isolated Iron Species



Isolated Rubber Material

A high paraffin wax content was observed, with the isolated paraffin exhibiting a melting point of 60–63 °C. Scale deposits composed of calcium carbonate and iron sulfide were also detected in the sample, with calcium carbonate confirmed by FTIR analysis. Additionally, rubber-like material was identified in the sample.

EILAND 34 1

Pump Serial: AS-1508
Model: 20-125-RHBC-20-5-4
Day Reported Down: 12/2/2025
Days Run: 517
Failure Cause: Tubing Failure

Inspection Summary: Inspection revealed the top plunger adapter was galled to the plunger. The plunger was worn to .005. All valve cages had exceeded the recommended service date. The barrel was bent and worn to .003.

Feet	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	
Prv BBL Out																																						
Barrel In	1	1	3	3	3	3	3	3	3	3	3	2	2	2	2	2	2	2	2	2	2																	
Barrel Out	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1																	
Prv PLG Out																																						
Plunger In	5	5	3	3	3	3																																
Plunger Out	5	5	5	5	5	5																																
Traveling Assembly		Partnumber					Description										Mtl Type		Part Condition		Qty		Replaced By															
Valve Rod Bushing		S27N-OSI					2-1/2" for 7/8" Valve Rod - Stainless, Bushing, Collet Type										SS		PART MISSING		1		S27N-OSI															
Collet Nut		S28N-OSI					2" & 2-1/2" Collet Nut-Stainless,Bushing										SS		PART MISSING		1		S28N-OSI															
Valve Rod		114K-OSI					7/8" Valve Rod Stock, Per Foot										CS		CUT		25		114K-OSI															
Top Plunger Adapter		M47D1-OSI					1-1/4" with 1/2" P.T. Monel,TPA										ML		GALLED		1		M47D1-OSI															
Plunger		102DQ5-5-M-G-OSI					1-1/4" X 5', -.005 FIT, PLUNGER,M.P. Grooved Body										SM/MP/G		WORN		1		102DQ5-5-M-G-OSI															
Traveign Cage		M39D-OSI					1-1/4" Closed API PIN END MNL										WLY		LIFE EXCEED		1		M39D-OSI															
Primary TV Ball		10DSN2-OSI					Ball, Silicon Nitrite - 11/16"										SN		LEAKED		1		10DSN2-OSI															
Primary TV Seat		11DC-OSI					Seat, Tungsten Carbide - 1-1/4"										CAR		LEAKED		1		11DC-OSI															
Seat Plug		S98D1-OSI					1-1/4" Recessed, Stainless, SEAT PLUG										SS		THREAD DAMAGE		1		S98D1-OSI															
Standing Assembly		Partnumber					Description										Mtl Type		Part Condition		Qty		Replaced By															
Valve Rod Guide		S70K1-V-OSI					2" for 7/8" Rod, Stainless - Vertical Discharge,GUIDE, ROD B										SS		GALLED		1		S70K1-V-OSI															
Barrel Bushing		S41K-M-OSI					Guide Mandrel, for Top Delrin Centralizer or Sand shield. 2"x15/16"										SS		GALLED		1		S41K-M-OSI															
Top Extension		B45D48-OSI					1-1/4" x 48" Brass,COUPLING, EXTENSION RH										BR/CID		THREAD DAMAGE		1		B45D48-OSI															
Barrel		B16D20N-OSI					20' Brass, Nickle Carbide,BARREL TH, 1-5/8" O.D. x 1-1/4" L										BR/CID		BENT		1		B16D20N-OSI															
Primary SV		S39H1-OSI					1-3/4" Modified, 1-1/4" RH Barrel, Stainless,CAGE, CLOSED PI										SS		LIFE EXCEED		1		S39H1-OSI															
Primary SV Ball		10HSN2-OSI					Ball, Silicon Nitride - 1"										SN		LEAKED		2		10HSN2-OSI															
Primary SV Seat		11HC-OSI					Seat, Tungsten Carbide - 1-3/4"										CAR		LEAKED		2		11HC-OSI															
DbI SV Cage		S38H-OSI					1-3/4" Stainless,CAGE, CLOSED DOUBLE VALVE										SS		LIFE EXCEED		1		S38H-OSI															
Hold Down Mandrel		S85K-OSI					2" API with 1.875 No Go, Stainless,MANDREL, API H.D., 3 CUP										SS		GALLED		1		S85N-OSI															
Spacer Rings		S110K-OSI					2" Type HR, Stainless, SPACER RING										SS		GALLED		2		S110N-OSI															
Seating cups		625J+30P-OSI					1-25/32" +30 #100, SEATING CUPS, TYPE "HR"										PLASTI		WORN		3		625J+30P-OSI															
Lock Nut		S95K-OSI					2" Stainless LOCK NUT,CUP HOLD DOWN										SS		GALLED		1		S95N-OSI															
Strainer Bushing		S50K-OSI					2" with 1" P.T. Stainless,COUPLING, HOLD DOWN										SS		GALLED		1		S50N-OSI															
Reason For Pulling: TUBING LEAK																		Foreign Materials:																				
Pump Desc (After): 20-125-RHBC-20-5-4																		Accessories: CVS-DBL																				
Deliver To: Stock																		Max Stroke (After): 214																				
Plunger Fit (After):																		Valve Rod Size: 7/8"																				
Remark/Notes: VALVE ROD BUSHING + COLLET NUT MISSING, VALVE ROD CUT ON LOCATION, TPA GALLED TO PLUNGER, PLUNGER WORN , ALL CAGES TRAVELLING + SV WERE ALL LITE EXCEED, ALL BALLS + SEATS LEAKED, VALVE ROD GUIDE + BUSHING BARREL WORN + BENT, ALL HOLD DOWN PARTS GALLED BUILT NEW PUMP W/NEW PUMP NUMBER																																						

EILAND 34 1



Plunger / Worn – TPA Galled



Exceeded recommended Run Life cage.

EILAND 34 1



Valve Rod was cut on location.

OSI Rod Pump Division - Failure Inspection

Inspection Summary: The pump arrived with the standing valve loose. Inspection revealed the valve rod bushing was galled to the collet nut, and the collet nut showed tagging damage. The valve rod exhibited pitting. The top plunger adapter and valve rod guide also showed tagging. Barrel wear measured to .003, and plunger mic readings measured .006, indicating notable wear. These conditions reflect mechanical wear and potential impact related damage affecting overall pump performance.

WEAR (THOUSANDTHS INCH AT FEET FROM TOP)																																						
Feet	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	
Prv BBL Out																																						
Barrel In	2	2	2	2	2	2	2	3	3	3	3	3	3	2	2	2	2	2	2	2	2	2	2	2														
Barrel Out	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1														
Prv PLG Out																																						
Plunger In	6	6	5	3	3	3																																
Plunger Out	5	5	5	5	5	5																																
Traveling Assembly			Partnumber						Description						Mtl Type		Part Condition		Qty		Replaced By																	
Valve Rod Bushing			BR22-1C						BUSHING COLLET875VR 1-1/16						SS		GALLED		1		S27N-OSI																	
Collet Nut			NC112						NUT COLLET LOCKNUT BUSHING						SS		POUNDED		1		S28N-OSI																	
Valve Rod			114K-OSI						7/8" Valve Rod Stock, Per Foot						CS		PITTED		25		114K-OSI																	
Top Plunger Adapter			M47D1-OSI						1-1/4" with 1/2" P.T. Monel,TPA						ML		POUNDED		1		M47D1-OSI																	
Plunger			102DQ5-3-M-OSI						1-1/4" X 5', -.003 FIT, PLUNGER,M.P.						SM/MIP		WORN		1		102DQ5-5-M-OSI																	
Traveign Cage			M39D-IG						CAGE CLOSED PIN END PLU						ML		WORN		1		M39D-OSI																	
Primary TV Ball			10DTC-OSI						Ball, Titanium Carbide - 3/4"						TIT		LEAKED		1		10DSN2-OSI																	
Primary TV Seat			11DC-OSI						Seat, Tungsten Carbide - 1-1/4"						CAR		LEAKED		1		11DC-OSI																	
Seat Plug			S98D1-OSI						1-1/4" Recessed, Stainless, SEAT PLUG						SS		GALLED		1		S98D1-OSI																	
Standing Assembly			Partnumber						Description						Mtl Type		Part Condition		Qty		Replaced By																	
Valve Rod Guide			S70K1-V-OSI						2" for 7/8" Rod, Stainless - Vertical Discharge,GUIDE, ROD B						SS		POUNDED		1		S70K1-V-OSI																	
Top Extension			B45D48-OSI						1-1/4" x 48" Brass,COUPLING, EXTENSION RH						BR		THREAD DAMAGE		1		B45D48-OSI																	
Barrel			B16D20N-OSI						20" Brass, Nickle Carbide,BARREL TH, 1-5/8" O.D. x 1-1/4" L						BR/CID		WORN		1		B16D20N-OSI																	
Bottom Extension			B45D03-OSI						1-1/4" x 3", Brass,COUPLING, EXTENSION RH						SS		THREAD DAMAGE		1		B45D03-OSI																	
Primary SV			CF471						1-3/4" CAGE BARREL 1.850						SS		WORN		1		S37H-OSI																	
Primary SV Ball			10HSN2-OSI						Ball, Silicon Nitride - 1"						SN		LEAKED		2		10HSN2-OSI																	
Primary SV Seat			11HC-OSI						Seat, Tungsten Carbide - 1-3/4"						CAR		LEAKED		2		11HC-OSI																	
Dbl SV Cage			S38H-OSI						1-3/4" Stainless,CAGE, CLOSED DOUBLE VALVE						SS		PART MISS		1		S38H-OSI																	
Hold Down Mandrel			S85N-OSI						2-1/2" API with 2.337 No Go, Stainless,MANDREL, API H.D., 3						SS		WORN		1		S85N-OSI																	
Spacer Rings			S110N-OSI						2-1/2" Type HR, Stainless, SPACER RING						SS		WORN		2		S110N-OSI																	
Seating cups			625M+70P-OSI						2-1/4" +70 #100 Plastic, SEATING CUPS, TYPE "HR"						PLASTI		WORN		3		625M+70P-OSI																	
Lock Nut			S95N-OSI						2-1/2" StainlessLOCK NUT,CUP HOLD DOWN						SS		WORN		1		S95N-OSI																	
Strainer Bushing			S50N-OSI						2-1/2" with 1-1/4" P.T. Stainless,COUPLING, HOLD DOWN						SS		GALLED		1		S50N-OSI																	
SV Misc			S118K1-OSI												SS																							
SV Misc			S118K2-OSI												SS																							
BARREL BUSHING			S41K-M-OSI						Guide Manndrel, for Top Delrin Centralizer or Sand shield. 2"x15/16"								GALLED		1		S41K-M-OSI																	
C-G-1			CG1-OSI						Delrin Centralizer Guide						NY		GALLED		1		CG1-OSI																	
Reason For Pulling: TUBING FAILURE																			Foreign Materials:																			
Pump Desc (After): 25-125-RHBC-24-5-4																			Accessories: CVS-DBL																			
Deliver To: Stock																			Max Stroke (After): 261																			
Plunger Fit (After): -5																			Valve Rod Size: 7/8"																			
Remark/Notes: VROD BUSHING GALLED,CUT COLLET NUT, POUNDED VROD PITTED , TPA POUNDED PLUNGER WORND TV CAGE WORN ALL BALLS + SEAT LEAKED VROD POUNDED BARREL WORN ALL HOLD DOWN PARTS WORN PUMP THAT CAME IN WAS ALL LOSE CAUSE OF TAGGING IN WELL MADE NEW PUMP DBO-29																																						

EILAND 34 2



EILAND 34 2



Lock Nut galled to the hold down.

EILAND 34 2



Thread damage and Exceeded recommended Run Life of cage.

EILAND 34 2



Worn plunger / TPA galled to plunger .

Pump Serial: DBO-31
Model: 25-150-RXBC-28-5-0
Day Reported Down: 12/2/2025
Days Run: 147
Failure Cause: Tubing Failure

Inspection Summary: Inspection revealed the barrel was worn. The valve rod showed signs of pitting, and the valve rod bushing was galled to the collet nut. The valve rod guide and barrel bushing were worn. All balls and seats did not hold vacuum during testing, indicating loss of sealing integrity.

WEAR (THOUSANDTHS INCH AT FEET FROM																																					
Feet	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
Prv BBL Out																																					
Barrel In	0	2	2	2	2	2	2	2	2	2	3		2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2									
Barrel Out	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1										
Prv PLG Out																																					
Plunger In	0	6	3	3	3	5																															
Plunger Out																																					
Traveling Assembly			Partnumber							Description							Mtl Type		Part Condition		Qty		Replaced By														
Valve Rod Bushing			S27N-OSI							2-1/2" for 7/8" Valve Rod - Stainless, Bushing, Collet Type							SS		GALLED		1		S27N-OSI														
Collet Nut			S28N-OSI							2" & 2-1/2" Collet Nut- Stainless,Bushing							SS		GALLED		1		S28N-OSI														
Valve Rod			114K-OSI							7/8" Valve Rod Stock, Per Foot							CS		PITTED		25		114K-OSI														
Top Plunger Adapter			M47D1-OSI							1-1/4" with 1/2" P.T. Monel,TPA							ML		GALLED		1		M47D1-OSI														
Plunger			102DQ5-5-M-G-OSI							1-1/4" X 5', -.005 FIT, PLUNGER,M.P. Grooved Body							SM/MP		WORN		1		102DQ5-5-M-G-OSI														
Traveign Cage			M39D-OSI							1-1/4" Closed API PIN END MNL							WLY		LIFE EXCEED		1		M39D-OSI														
Primary TV Ball			10DSN2-OSI							Ball, Silicon Nitrite - 11/16"							SN		WORN		1		10DSN2-OSI														
Primary TV Seat			11DC-OSI							Seat, Tungsten Carbide - 1-1/4"							CAR		WORN		1		11DC-OSI														
Seat Plug			S98D1-OSI							1-1/4" Recessed, Stainless, SEAT PLUG							SS		GALLED		1		S98D1-OSI														
Standing Assembly			Partnumber							Description							Mtl Type		Part Condition		Qty		Replaced By														
Valve Rod Guide			S70K1-V-OSI							2" for 7/8" Rod, Stainless - Vertical Discharge,GUIDE, ROD B							SS		WORN		1		S70K1-V-OSI														
Barrel Bushing			S41K-M-OSI							Guide Mandrel, for Top Delrin Centralizer or Sand sheild. 2"x15/16"							SS		WORN		1		S41K-M-OSI														
Top Extension			B45D24-OSI							1-1/4" x 24" Brass,COUPLING, EXTENSION RH							BR/CID		AAD		1		B45D24-OSI														
Barrel			B16D26N-OSI							26" Brass, Nickle Carbide,BARREL TH, 1-5/8" O.D. x 1-1/4" I.D.							BR/CID		WORN		1		B16D26N-OSI														
Primary SV			S39H1-OSI							1-3/4" Modified, 1-1/4" RH Barrel, Stainless,CAGE, CLOSED PI							SS		LIFE EXCEED		1		S39H1-OSI														
Primary SV Ball			10HSN2-OSI							Ball, Silicon Nitride - 1"							SN		WORN		2		10HSN2-OSI														
Primary SV Seat			11HC-OSI							Seat, Tungsten Carbide - 1-3/4"							CAR		WORN		2		11HC-OSI														
Dbt SV Cage			S38H-OSI							1-3/4" Stainless,CAGE, CLOSED DOUBLE VALVE							SS		LIFE EXCEED		1		S38H-OSI														
Hold Down Mandrel			S85K-OSI							2" API with 1.875 No Go, Stainless,MANDREL, API H.D., 3 CUP							SS		WORN		1		S85K-OSI														
Spacer Rings			S110K-OSI							2" Type HR, Stainless, SPACER RING							SS		WORN		2		S110K-OSI														
Seating cups			625J+30P-OSI							1-25/32" +30 #100, SEATING CUPS, TYPE "HR"							PLASTI		WORN		3		625J+30P-OSI														
Lock Nut			S95K-OSI							2" Stainless LOCK NUT,CUP HOLD DOWN							SS		GALLED		1		S95K-OSI														
Strainer Bushing			S50K-OSI							2" with 1" P.T. Stainless,COUPLING, HOLD DOWN							SS		GALLED		1		S50K-OSI														
C-G-1			CG1-OSI							Delrin Centralizer Guide							NY		WORN		1		CG1-OSI														
ADAT			S24KN1-OSI							2"pump for 2 1/2" Hold Down Stainless Bushing Adapter HD							SS		GALLED		1		S24KN1-OSI														
Reason For Pulling: TUBING FAILURE																			Foreign Materials:																		
Pump Desc (After): 25-125-RHBC-26-5-2																			Accessories: CVS-DBL																		
Deliver To: Stock																			Max Stroke (After): 26																		
Plunger Fit (After):																			Valve Rod Size: 7/8"																		
Remark/Notes: CONVERTED PUMP TO SPECIFICATIONFOR WELL BARREL WN PLUNGER WN VALVE ROD PIT VALVE ROD BUSHING GAL COLLET NUT GAL PRIMARY TV BALL AND SEA WN SEAT PLUG GAL VALVE ROD GUIDE WN BARREL BUSHING WN TOP EXT ADD PRIMARY SV CAGE LIFE PRIMARY SV BALL AND SEAT WN DBL SV CAGE LIFE HOLDDOWN MANDREL WN																																					

HARMONSON - 11 -1



Plunger/ grooved / worn.



Tagged, valve rod
guide.

HARMONSON - 11 -1



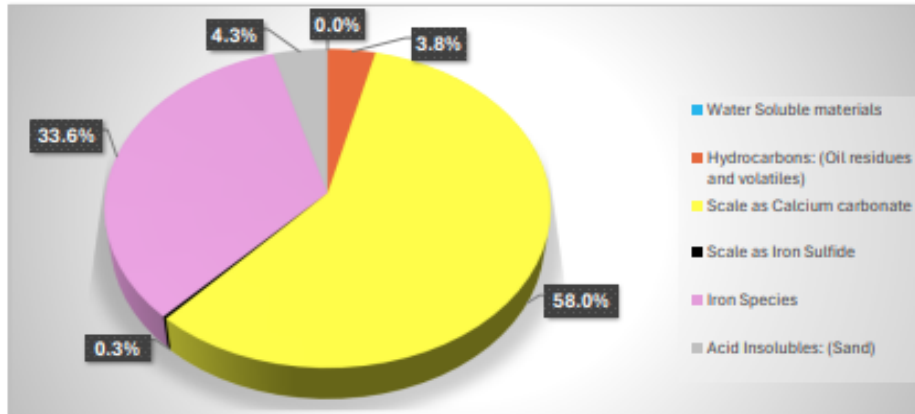
Pitted Valve Rod.

HARMONSON - 11 -1

SOLID COMPOSITION

Water Soluble materials	0.0%
Hydrocarbons: (Oil residues and volatiles)	3.8%
Scale as Calcium carbonate	58.0%
Scale as Iron Sulfide	0.3%
Iron Species	33.6%
Acid Insolubles: (Sand)	4.3%

Solid Appearance: **Dark gray solid**



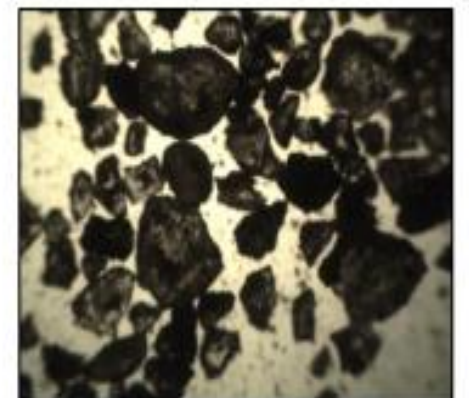
SOLID APPEARANCE



Initial Sample



Isolated Iron Species



After Cleaning (Microscopic View)

A high quantity of scale deposits composed of calcium carbonate was detected in the sample, with its presence confirmed by FTIR analysis. Traces of iron sulfide scale were also detected. Additionally, a significant amount of iron species was observed, likely resulting from a corrosive environment.